Econ 812 Recitation 5 Notes

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1 Two quick Log Linearization examples

Follow these steps to log linearize:

Step 1 Take derivative of each side and evaluate at steady state

Step 2 Multiply by steady state variable

Step 3 Multiply by log-linearized variable

Step 4 Continue to next variable

Step 5 Simplify

1.1 From class

$$y_t = k_t^{\alpha} l_t^{1-\alpha}$$

$$1 * y * \hat{y_t} = \alpha k^{\alpha-1} l^{1-\alpha} * k * \hat{k_t} + (1-\alpha) k^{\alpha} l^{-\alpha} * l * \hat{l_t}$$

knowing that $y = k^{\alpha} l^{1-\alpha}$ we have

$$k^{\alpha} l^{1-\alpha} * \hat{y_t} = \alpha k^{\alpha} l^{1-\alpha} * \hat{k_t} + (1-\alpha) k^{\alpha} l^{1-\alpha} * \hat{l_t}$$

$$\hat{y_t} = \alpha \hat{k_t} + (1-\alpha) \hat{l_t}$$

1.2 From old comps

$$c_t n_t = (1-\tau_t) w_t$$

$$n * c * \hat{c_t} + c * n * \hat{n_t} = (1-\tau) * w * \hat{w_t} + -w * \tau * \hat{\tau_t}$$

$$cn(\hat{c_t} + \hat{n_t}) = (1-\tau) w \hat{w_t} - w\tau\hat{\tau_t}$$

$$\hat{c_t} + \hat{n_t} = \hat{w_t} - \frac{\tau}{1-\tau}\hat{\tau_t}$$

2 Fall 22 Midterm Question 2

Consider the following problem:

$$\max \sum_{t=0}^{\infty} \beta^t z_t \ln c_t \tag{1}$$

by choice of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ subject to

$$c_t + k_{t+1} = 2k_t^{1/2} \tag{2}$$

$c_t, k_{t+1} \geq 0$ and $k_0 > 0$ and $\{z_t\}$ given and $\beta = 2/3$

Note: Remember the steps for solving the Lagrangean!

2.1 Steps to solving a Lagrangian DP problem

1. Set up Lagrangian, which has form of objective function + constraints*multiplier
2. Take FOCs
3. Combine FOCs to get Euler Equation (first order difference equation)
4. Solve for policy functions

2.2 Part A

"Set up the Lagrangean for this problem, state the first-order conditions and the transversality condition (you may ignore the non-negativity constraints on k,c,z)"

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [z_t \ln c_t - \lambda_t (c_t + k_{t+1} - 2k_t^{1/2}) \tag{3}$$

subject to $c_t + k_{t+1} = 2k_t^{1/2}$

Take FOC's

$$c_t : \frac{\beta^t z_t}{c_t} - \beta^t \lambda_t = 0 \tag{4}$$

$$k_{t+1} : -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} k_{t+1}^{-1/2} = 0 \tag{5}$$

$$\lambda_t : c_t + k_{t+1} = 2k_t^{1/2} \tag{6}$$

TVC: $\lim_{t \to \infty} \beta^t \lambda_t k_{t+1} = 0$

2.3 Part B

"Eliminate the multiplier on the constraint from your first order conditions to get a two dimensional first order difference equation in c and k."

Combine equations 4 and 5 to get the first dimension

$$\frac{z_t}{c_t} = \beta \frac{z_{t+1}}{c_{t+1}} k_{t+1}^{-1/2} \tag{7}$$

The second dimension is the resource constraint

$$c_t + k_{t+1} = 2k_t^{1/2} \tag{8}$$

2.4 Part C

"Suppose that $z_t = 1$ for all t. Solve for the steady state solution $c_t = c > 0$ and $k_t = k > 0$"

$$\frac{1}{c} = 2/3 \frac{1}{c} k^{-1/2} \tag{9}$$

$$k = (2/3)^2 = 4/9 \tag{10}$$

$$c + 4/9 = 2(4/9)^{1/2} \tag{11}$$

$$c = 2\beta - \beta^2 = 4/3 - 4/9 = 8/9 \tag{12}$$

2.5 Part D

Log linearize the non-linear difference equations around the steady state and express the system assume

$$A \begin{bmatrix} \hat{k_{t+1}} \\ \hat{c_{t+1}} \end{bmatrix} = B \begin{bmatrix} \hat{k_t} \\ \hat{c_t} \end{bmatrix} + C \begin{bmatrix} \hat{z_{t+1}} \\ \hat{z_t} \end{bmatrix}$$

where A,B and C are 2 x 2 matrices with standard notation

$$\hat{x_t} := \frac{x_t - x}{x}$$

where x is a steady state value for x_t. Use $z = 1$ for the steady state value of z_t.

Our two dimensional first order difference equation:

$$\frac{z_t}{c_t} = \beta \frac{z_{t+1}}{c_{t+1}} k_{t+1}^{-1/2} \tag{13}$$

$$c_t + k_{t+1} = 2k_t^{1/2} \tag{14}$$

To log linearize, follow these steps:

Step 1 Take derivative of each side and evaluate at steady state

Step 2 Multiply by steady state variable

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Step 4 Continue to next variable

Step 5 Simplify

For equation 1:

$$\frac{1}{c}*z*\hat{z}_t+\frac{-z}{c^2}*c*\hat{c}_t = \beta\frac{1}{c}k^{-1/2}*z*\hat{z}_{t+1}+\beta z(-1/2)k^{-3/2}\frac{1}{c}*k*\hat{k}_{t+1}+\beta z k^{-1/2}\frac{-1}{c^2}*c*\hat{c}_{t+1}$$

Simplifying...

$$\frac{z}{c}(\hat{z}_t-\hat{c}_t) = \beta\frac{z}{c}k^{-1/2}(\hat{z}_{t+1}-1/2\hat{k}_{t+1}-\hat{c}_{t+1})$$

$$1/2\hat{k}_{t+1}+\hat{c}_{t+1} = \hat{z}_{t+1}-\hat{z}_t+\hat{c}_t$$

For equation 2:

$$1*c*\hat{c}_t+1*k*\hat{k}_{t+1} = k^{-1/2}*k*\hat{k}_t$$

$$k\hat{k}_{t+1} = k^{1/2}\hat{k}_t-c\hat{c}_t$$

Now we want to line these up to fit with the coefficient matrices.

$$A = \begin{bmatrix} 1/2 & 1 \\ k & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 \\ k^{1/2} & -c \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

And we know from the previous part what k and c are.

3 Solution to PS3

3.1 Question 1 Part A

$$\mathscr{L} = \sum_t \sum_{\epsilon^t} \beta^t \Pi(\epsilon^t)[\epsilon_t \log c(\epsilon^t)+\lambda(\epsilon^t)[W(\epsilon^{t-1}-W(\epsilon^t)-c(\epsilon^t)+\mu(\epsilon^T)W(\epsilon^T))]]$$

FOC for $c(\epsilon^t)$

$$c(\epsilon^t) : \beta^t \sum_{\epsilon^t} \Pi(\epsilon^t)\epsilon_t(1/c(\epsilon^t)) = \beta^t \sum_{\epsilon^t} \Pi(\epsilon^t)\lambda(\epsilon^t)$$

This implies that

$$\epsilon_t/c(\epsilon^t) = \lambda(\epsilon^t)$$

FOC for $W(\epsilon^{t+1})$

$$W(\epsilon^{t+1}) : \beta^t \sum_{\epsilon^t} \Pi(\epsilon^t)\lambda(\epsilon^t) = \beta^{t+1} \sum_{\epsilon^{t+1}} \Pi(\epsilon^{t+1})\lambda(\epsilon^{t+1})$$

This implies that

$$\lambda(\epsilon^t) = \beta E[\lambda(\epsilon^{t+1})|\epsilon^t]$$

FOC for $W(\epsilon^T)$

$$W(\epsilon^T) : \lambda(\epsilon^T) = \mu(\epsilon^T)$$

TVC:

$$\mu(\epsilon^T)W(\epsilon^T) = 0$$

Plus the resource constraint

3.2 Part B

This is a concave programming problem with linear constraints

3.3 Part C

Solving for the EE

$$\frac{\epsilon_t}{W(\epsilon^{t-1}) - W(\epsilon^t)} = E[\frac{\beta\epsilon_{t+1}}{W(\epsilon^t) - W(\epsilon^{t+1})}|\epsilon_t]$$

Define as before $z(\epsilon^t) = \frac{W(\epsilon^{t+1})}{W(\epsilon^t)}$. Substitute to find

$$\frac{\epsilon_t z(\epsilon^t)}{1 - z(\epsilon^t)} = E[\frac{\beta\epsilon_{t+1}}{1 - z(\epsilon^{t+1})}|\epsilon_t]$$

With TVC $z(\epsilon^T) = 0$

3.4 Part D

We have from the probability matrix and unconditional probability for ϵ_t of

$$E[\epsilon_t] = 1.25$$

And we have that $z_T = 0$. From the equation above we have:

$$\frac{\epsilon_{T-1} z(\epsilon^{T-1})}{1 - z(\epsilon^{T-1})} = E[\beta * 1.25]$$

Rearrange to find

$$z_{T-1} = \frac{\beta * 1.25}{\epsilon_{T-1} + \beta * 1.25}$$

Plugging back into the general solution, we find

$$\frac{\epsilon_{T-2} z(\epsilon^{T-2})}{1 - z(\epsilon^{T-2})} = E[\frac{\beta \epsilon_{T-1}}{1 - \frac{\beta * 1.25}{\epsilon_{T-1} + \beta * 1.25}} | \epsilon_t]$$

$$z_{T-2} = \frac{(\beta + \beta^2)1.25}{\epsilon_{T-2} + (\beta + \beta^2)1.25}$$

So the general solution is

$$z_{T-j} = \frac{\sum_{j=1}^{20} \beta^j 1.25}{\epsilon_{T-j} + \sum_{j=1}^{20} \beta^j 1.25}$$

3.5 Question 2 Part A

The Bellman Equation here is:

$$V(W_{t-1}, \epsilon_t) = \max \epsilon_t \log c_t + E[\beta V(W_t, \epsilon_{t+1}) | \epsilon_t]$$

With state variables W_{t-1}, ϵ_t and control variable c_t

Part B

In the infinite case, replace the sums with the solution to the infinite geometric sum:

$$z(\epsilon) = \frac{1.25\beta/(1-\beta)}{\epsilon + 1.25\beta/(1-\beta)}$$

We already know the policy functions from the previous problem: $c = (1 - z)W$ and $W' = zW$. So plugging into the Bellman:

$$V(W, \epsilon) = \epsilon[\log((1-z)W)] + E[\beta V(z(\epsilon)W, \epsilon')]$$

With the guess of $V(W, \epsilon) = A(\epsilon) + B(\epsilon) \log W$
Expanding out we have

$$\epsilon \log(1-z)W + \beta E[A(\epsilon') + B(\epsilon') \log z + B(\epsilon') \log W]$$

Matching up coefficients we have

$$B(\epsilon) = \epsilon + \beta E[B(\epsilon')]$$

Then we also have $E[B(\epsilon)] = E[B(\epsilon')] = 1.25 + \beta E[B(\epsilon')]$. So simplifying we find that $E[B(\epsilon')] = \frac{1.25}{1-\beta}$. Then replacing we find

$$B(\epsilon) = \epsilon + \frac{1.25\beta}{1-\beta}$$

And we can collect the remaining terms for $A(\epsilon)$